

```
#include <iostream>

#include <chrono>

using namespace std;

// Iterative Method
// Time Complexity: O(n)
// Space Complexity: O(1)
unsigned long long iterativeFactorial(int number)
{
    unsigned long long factorial = 1;

    for (int count = 2; count <= number; count++)
    {
        factorial *= count;
    }

    return factorial;
}

// Recursive Method
// Time Complexity: O(n)
// Space Complexity: O(n) because of recursive calls
unsigned long long recursiveFactorial(int number)
{
    if (number == 0 || number == 1)
        return 1;

    return number * recursiveFactorial(number - 1);
}
```

```
}
```

```
int main()
```

```
{
```

```
    int value;
```

```
    cout << "Enter a non-negative integer: ";
```

```
    if (!(cin >> value) || value < 0)
```

```
    {
```

```
        cout << "Error: Please enter a valid non-negative integer." << endl;
```

```
        return 1;
```

```
    }
```

```
    // Measure execution time for Iterative Method
```

```
    auto iterativeStart = chrono::high_resolution_clock::now();
```

```
    unsigned long long iterativeAnswer = iterativeFactorial(value);
```

```
    auto iterativeEnd = chrono::high_resolution_clock::now();
```

```
    chrono::duration<double, nano> iterativeTime = iterativeEnd - iterativeStart;
```

```
    // Measure execution time for Recursive Method
```

```
    auto recursiveStart = chrono::high_resolution_clock::now();
```

```
    unsigned long long recursiveAnswer = recursiveFactorial(value);
```

```
    auto recursiveEnd = chrono::high_resolution_clock::now();
```

```
    chrono::duration<double, nano> recursiveTime = recursiveEnd - recursiveStart;
```

```
    // Display Results
```

```
    cout << "\n===== Factorial Results =====" << endl;
```

```
cout << "Input Number    : " << value << endl;
cout << "\nIterative Method" << endl;
cout << "Factorial        : " << iterativeAnswer << endl;
cout << "Execution Time    : " << iterativeTime.count() << " ns" << endl;

cout << "\nRecursive Method" << endl;
cout << "Factorial        : " << recursiveAnswer << endl;
cout << "Execution Time    : " << recursiveTime.count() << " ns" << endl;

return 0;
}
```